Botany short course 2026

Leaves

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What is a leaf?

- Variable in size and shape, form and function

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- Photosynthetic, protective (spines/sheaths), architectural (climbing), carnivorous (pitchers), storage (e.g. garlic)

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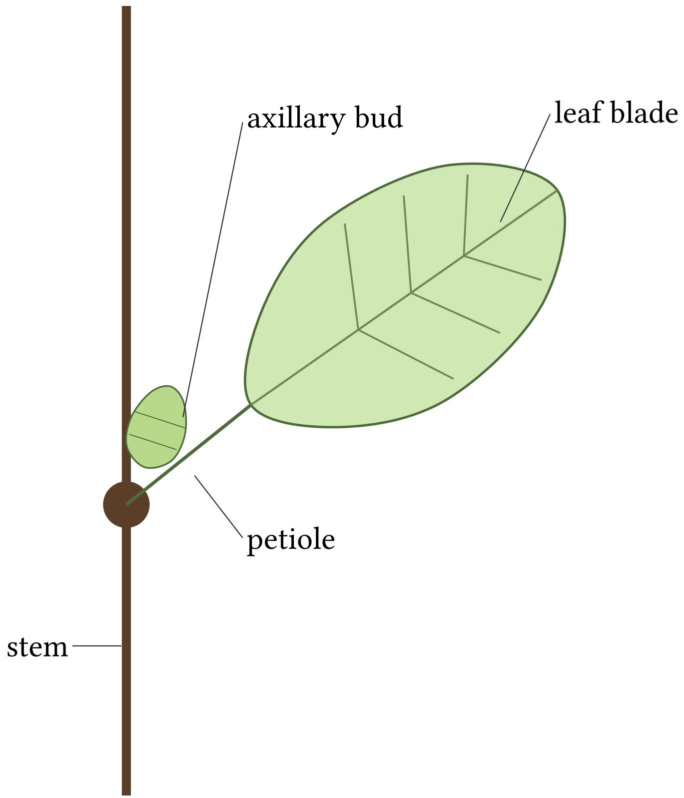
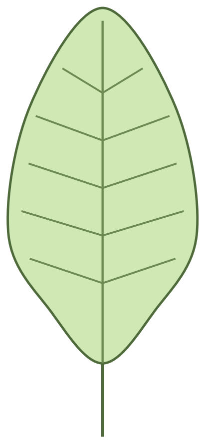
- Variable in size and shape, form and function
- Photosynthetic, protective (spines/sheaths), architectural (climbing), carnivorous (pitchers), storage (e.g. garlic)
- **A leaf always subtends a bud**

Leaves always subtend a bud

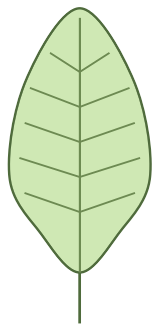


Leaf characters you will classify

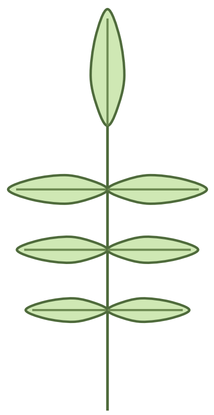
- Simple and compound
- Margins
- Bases
- Overall shape
- Venation
- Any hairs?



Simple vs compound



Simple



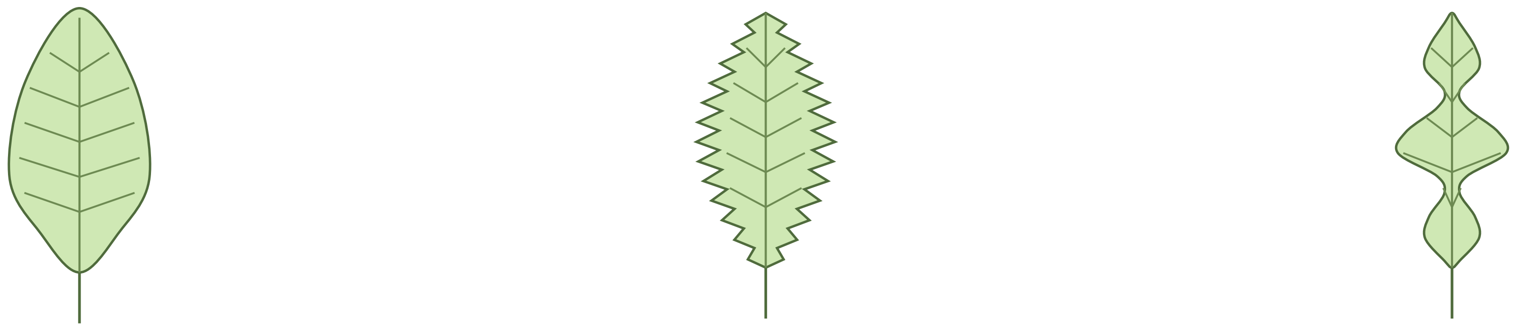
Compound: 1-pinnate

Compound: palmate

Margins and bases

- Margins look at leaf edges
- Bases are where the petiole joins the leaf

Leaf margins

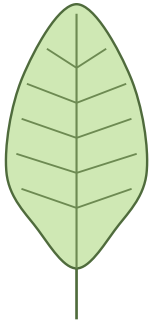


Entire

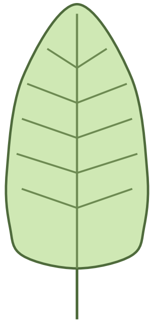
Toothed

Lobed

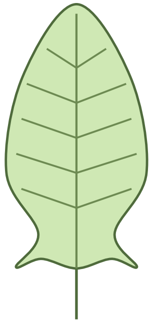
Leaf bases



Cuneate



Rounded

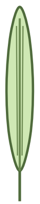


Hastate / sagittate

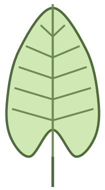
Shape and venation

- Overall leaf shape. There are **many**, we only show a few...
- What do the leaf **veins** look like

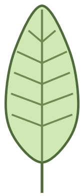
Overall leaf shape



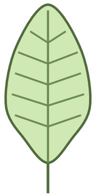
Linear



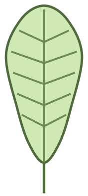
Cordate



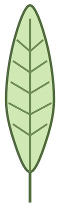
Elliptic



Ovate

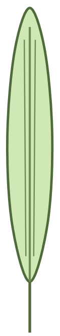


Obovate

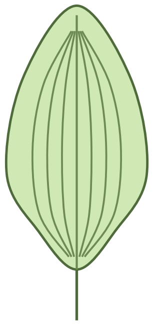


Lanceolate

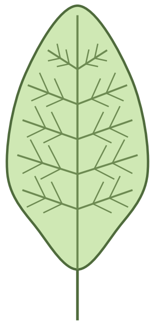
Venation



Parallel (grass-like)



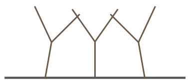
Parallel (*Plantago*)



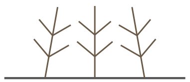
Net-veined

Leaf hairs

Simple hairs (many species)

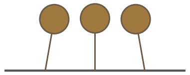


Forked hairs (Brassicaceae)

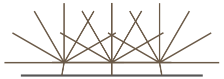


Dendritic hairs (e.g. *Verbascum*)

Hooked hairs (e.g. *Myosotis*)

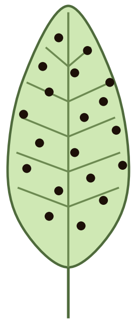


Glandular hairs (many species)

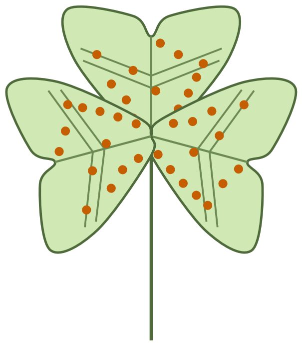


Stellate hairs (e.g. *Hedera*)

Leaf glands



Black scattered glands (e.g. *Hypericum*)



Orange scattered glands (e.g. *Oxalis*)

Exercise

- There are leaves from different species
- Take one example from each species
- Group similar leaves together
- Describe each group of leaves as best you can